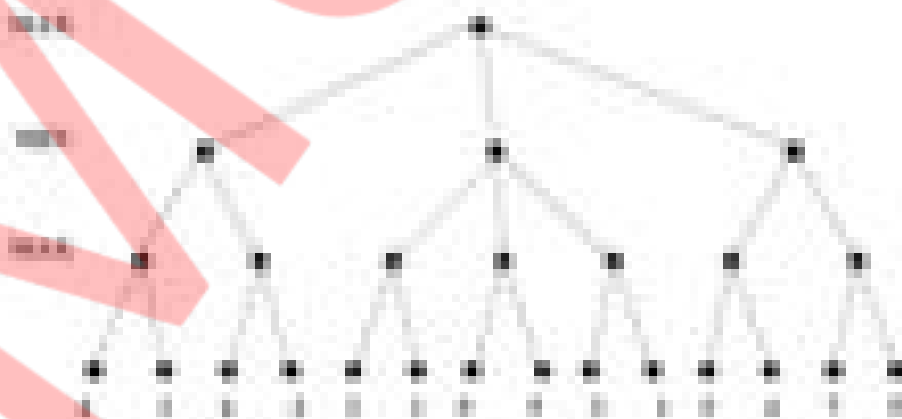


Time: 3 Hours

[Total Marks: 80]

Note: i) Question no. 1 is compulsory
 ii) Attempt any three from remaining
 iii) Assume necessary data

- 1 Solve any four of the following:
- (a) Differentiate between Model-Based Agents and Goal-Based Agents. 5
 - (b) Compare A* search with Greedy Best-First Search 5
 - (c) Give the PEAS and State-Space description for an Interactive English Tutor. 5
 - (d) Give the types of Quantifiers. Translate the following sentences into First-Order Predicate Logic (FOPL) using quantifiers. 5
 - i) Every student who studies hard passes the exam.
 - ii) Some doctors are researchers.
 - iii) All humans are mortal.
 - iv) There exists a country where everyone loves cricket.
 - (e) Explain various methods of knowledge representation. 5
- 2
- (a) Discuss the various properties of task environments in detail. Illustrate with an example of an Autonomous Delivery Drone. 10
 - (b) Apply alpha beta pruning tree on following graph 10



- 3
 - (a) Illustrate forward-chaining algorithm with suitable example. 10
 - (b) Explain the Simulated Annealing Algorithm. Compare it with Hill Climbing. Show with an example how annealing helps escape local maxima 10
- 4
 - (a) Explain the working of Unification and Resolution in FOPL. Prove by resolution:
'If Anil eats peanuts and Harry eats everything that Anil eats, then Harry eats peanuts.' 10
 - (b) Explain the STRIPS representation for planning. Give an example problem and show its STRIPS formulation. 10
- 5
 - (a) Explain Hierarchical Planning with a suitable example. How is it different from Partial-Order Planning?. 10
 - (b) Define Belief Network. Describe the steps of constructing belief network with an example. 10
- 6

Write short notes on any Two of following:

 - (a) Explain three modern AI applications in Robotics 10
 - (b) Explain the role of Genetic Algorithms in optimization 10
 - (c) Wumpus world Environment 10

Duration: 3hrs

[Max Marks:80]

N.B. : (1) Question No 1 is Compulsory.

(2) Attempt any three questions out of the remaining five.

(3) All questions carry equal marks.

(4) Assume suitable data, if required and state it clearly.

- | | | |
|----------|--|-------------|
| 1 | Attempt any FOUR | [20] |
| a | How we can resolve the Hidden Station and Exposed Station Problems? Explain in short. | 5 |
| b | What are various Handover Mechanisms describe in brief. | 5 |
| c | What are various Differences between Infrastructure Network and Ad-hoc Networks? | 5 |
| d | Write a brief Note on Frequency Hopping Spread Spectrum. | 5 |
| e | List Various Wireless LAN Threats. Give description in short for each threat. | 5 |
| 2 | a Explain GPRS System Architecture in detail. | [10] |
| | b Explain Various Multiplexing Schemes that are beneficial in Communication. | [10] |
| 3 | a Explain GSM Architecture in Details. Describe all interfaces of GSM System. | [10] |
| | b Write a detailed comparison on Various Generations of Telecommunication. | [10] |
| 4 | a What is significance of A3 A5 and A8 Algorithm. Explain in detail with neat sketch. | [10] |
| | b Explain Tunneling and Encapsulation in detail. Describe all encapsulation schemes in brief. | [10] |
| 5 | a Explain the following Terms in brief:
i) Cellular Network ii) Snooping TCP. | [10] |
| | b Explain Mac management in IEEE 802.11. | [10] |
| 6 | a Explain IP Packet Delivery to and from Mobile Node? | [10] |
| | b What is Blue-tooth? Explain Bluetooth Architecture with neat Sketch. | [10] |

Time: 3 Hours

Total Marks: 80

- N.B:** (1) Question No. 1 is compulsory.
 (2) Attempt any three questions out of the remaining five questions.
 (3) Figures to the right indicate full marks.
 (4) Make suitable assumptions wherever necessary.

- Q.1.** A. Explain the two-layered model of system software. Justify the need for machine-dependent and machine-independent system programs. 5
 B. Construct operator precedence Parser for the grammar:

$$E \rightarrow E + E \mid E * E \mid a.$$
 Parse the string "a+a*a" using the same parser. 5
 C. Define Relocation and Linking. Why are they crucial functions of a linking loader 5
 D. Define Ambiguous Grammar. How is ambiguity a problem in the design of a parser? 5
- Q.2.** A. With reference to the LR parsing family, differentiate between SLR (1), CLR (1), and LALR (1) parsers based on the size of the parsing table and the power of the grammar they can handle. 10
 B. Construct Three address code for the following program 10
- ```

i = 1;
x = 0;
while (i <= n)
{
 x = x + 1;
 i = i + 1;
}

```
- Q.3.** A. Explain the various data structures used in the Two-Pass Assembler design, specifically focusing on the Opcode Table, Symbol Table, and Literal Table. 10  
 B. Explain the various features of a Macro facility, such as nested macro calls and conditional macro expansion, with suitable examples. 10
- Q.4.** A. Explain the working of a Single-pass macro processor with neat flowchart. 10  
 B. What are the different ways of representing Intermediate code? Explain with suitable example. 10
- Q.5.** A. Explain any five code optimization techniques in compiler designing with suitable example 10  
 B. Explain design issues in code generation in detail. 10
- Q.6.** A. Differentiate between Machine-Dependent and Machine-Independent code optimization techniques. Give two distinct examples for each category. 10  
 B. Explain advanced macro facilities with suitable examples. 10

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(3 Hours)

NB

- 1) Question **number 1** is compulsory
- 2) Attempt **any three** out of the remaining five questions.
- 3) Assume suitable data if **necessary** and justify the assumptions.
- 4) **Draw** neat and **clean** diagrams
- 5) Figures to the **right** indicate full marks

- Q1 Attempt the following 20
- a) Define Statistics. Explain any two functions of Statistics with examples.
  - b) Parametric point Estimation.
  - c) Explain different non-probability sampling techniques with examples
  - d) For the following raw data, prepare a frequency distribution with the starting class as 5–9 and all classes having the same width of 5.  
Marks in English:  
12, 36, 40, 16, 10, 10, 19, 20, 28, 30,  
19, 27, 15, 21, 33, 45, 7, 19, 20, 26,  
26, 37, 6, 5, 20, 30, 37, 17, 11, 20
- Q2 a) Calculate the Karl Pearson correlation coefficient for following data ,and comment on the relationship between two variables 10
- |    |    |    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|----|----|
| 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 |
| 38 | 36 | 34 | 32 | 30 | 28 | 26 | 24 | 22 | 20 |
- b) Prepare a suitable table showing the distribution of candidates, 10  
Out of a total number of 10,000 candidates who applied for jobs in a government department, 6,854 were males, 3,146 were graduates and the others were non-graduates. The number of candidates with some experience was 2,623, of whom 1,860 were males. The number of male graduates was 2,012. The number of graduates with experience was 1,093, which includes 323 females.
- Q3 a) A regression model predicts the following values: [4.1, 5.2, 6.9, 9.8, 14.5]. Actual values are: [4, 5, 7, 10, 15]. Calculate the coefficient of determination ( $R^2$ ) and interpret its meaning. 10
- b) State and explain the Neyman-Pearson Lemma. What is its significance in hypothesis testing? 10

- Q4 a) The following data shows the fuel efficiency of cars (Y in km/litre) based on engine size ( $X_1$  in cc) and vehicle weight ( $X_2$  in kg), frame regression equation and interpret the results. 10
- | Y  | $X_1$ | $X_2$ |
|----|-------|-------|
| 18 | 1200  | 900   |
| 15 | 1500  | 1100  |
| 14 | 1600  | 1200  |
| 12 | 1800  | 1300  |
| 10 | 2000  | 1400  |
- b) A marketing firm is launching a new energy drink targeted at young adults. They want to understand consumer behaviour, preferences, and price expectations before launching. 10
- Identify the type of data the firm should collect and explain why.
  - Suggest and explain two suitable methods for collecting this data.
  - Recommend two possible sources of secondary data that might also help in their research.
- Q5 a) From a regression output:  $Y=3+2X_1+0.8X_2$  10
- The standard errors of  $\beta_1$  and  $\beta_2$  are 0.6 and 0.3, respectively. Test whether each regression coefficient is significantly different from 0 at the 5% significance level. Use the t-test. At the **5% significance level (two-tailed)**, **critical value**  $\approx \pm 2.0$  (for large df).
- b) Explain in detail Null and Alternative Hypothesis with an example. 10
- Q6 Write short notes on 20
- Describe the types of statistical data with suitable examples.
  - Type I and Type II errors in hypothesis testing.
  - $R^2$ , MAE, MSE and MAPE.
  - Method of Maximum likelihood.

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**Duration: 3hrs**

**Max Marks: 80**

- N.B. : (1) Question No 1 is Compulsory.  
(2) Attempt any three questions out of the remaining five.  
(3) All questions carry equal marks.  
(4) Assume suitable data, if required and state it clearly.

- 1 Attempt any FOUR [20]
  - a Give examples of replay attacks. List three general approaches for dealing with replay attacks.
  - b Compare and contrast keyed and keyless transposition ciphers.
  - c Write short note on packet sniffing attack.
  - d Explain transposition cipher with example.
  - e What are computer viruses and worms? Write any four key differences between them.
- 2 a Find the shared secret key in a Diffie-Hellman scheme with a common prime 71 and primitive root 7. Given the private keys of Alice and Bob are 5 and 12 respectively. [10]  
b Explain AES. Compare and contrast AES vs DES. [10]
- 3 a What are different types of firewalls? How a firewall is different from an IDS. [10]  
b Explain layer wise TCP/IP Vulnerability. [10]
- 4 a Why are digital certificates and signatures required? What is the role of digital signature in digital certificates? Explain any one digital signature algorithm. [10]  
b How does PGP achieve confidentiality and authentication in emails? [10]
- 5 a Explain how you would use the MD5 hashing algorithm to verify the integrity of the received message. Show all the steps of the algorithm with the help of a diagram. [10]  
b Explain Kerberos. Why is it called an SSO? [10]
- 6 a Enlist the various functions of the different protocols of SSL. Illustrate the connection establishment between client and server in SSL handshake protocol in detail. [10]  
b What is Digital Signature? Explain RSA as a Digital Signature Scheme. [10]

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